

Much Thinkness 2026

a_person

1 Contest

1. Speedrun P1–2 in $< 15\text{m}$.
2. Solve P3 in $< 30\text{m}$.
3. Look at P4–6, solve at least one in $< 1\text{h}$.
4. Grind subtasks for the rest.
5. Vectorise every single suboptimal solution near the end.

2 Problem

1. Look at the subtasks (for P3–6).
2. Make some observations, do some cases by hand, and make some conjectures.
3. Analyse the bounds for the subtasks; see what is intended and what can fit.
4. Once you code early subtasks with easy observations, try to combine them into a full solution or brute force.
5. Change brute force into something better with some sort of DSA.

3 Subtasks

1. Always do the first few; they are quite easy and usually free points.
2. Check constraints, special properties, and do some cases by hand and generalise.
3. If needed, represent the problem mathematically, or with a graph or some other visualisation.
4. After making observations, you should be able to make some brute force or solve early subtasks.
5. Check how subtasks combine together, and try to make a full solve.

4 Problems

1. Except for baubles, should be extremely trivial.
2. Make some small observation and use simple DS.
3. Do subtasks quickly, make observations, and topic search.

4. Usually involves some annoying greedy or observation, but try to finish it kinda quickly with a similar strategy to P4. Make your mind wander more and have breaks/walks etc. Subtasks could get annoying, so be careful and pseudocode by hand.
5. Just grind these out of the subtasks and, yeah, make a good solve. Be very careful with implementation.
6. Realistically shouldn't be getting an AC on this. Just do lots of subtasks, and even if it seems trivial, **MAKE SURE TO WRITE PSEUDOCODE** to make sure implementation will not be super annoying.

5 Limits

$$10^1 : \quad N! \quad (N!N^2, 3^N N^3)$$

$$10^2 : \quad N^3 \quad (N^4, N^3 \log^2 N)$$

$$10^3 : \quad N^2 \quad (N^2 \log N)$$

$$10^4 : \quad N\sqrt{N} \quad (N^2)$$

$$10^5 : \quad N \log N \quad (N \log^2 N)$$

$$10^6 : \quad \alpha N \quad (N \log N)$$

$$10^7 : \quad N \quad (N \log N, \alpha N)$$

5.1 Memory Limits

- **8 MB:** amt is being bull and wants you to implement a super low-memory algorithm.
- **16 MB:** Stick to arrays and bitsets; no fancy segtree or even map.
- **32 MB:** A bit more better, but just if you don't bash or waste data, you're fine.
- **64 MB:** Don't bash and you're fine, unless constraints are quite high.
- **128 MB:** You can use loads of memory; don't really care about this. However, superbash might not work.
- **256 MB:** Bashing is fine; do whatever you want except being silly.
- **512 MB:** Isn't really a difference for C++ between 512 MB and 1 GB.
- **1 GB:** Do whatever you want.

6 Ideas

Think like the problemsetters:

- What would be a nice observation/greedy/etc.?
- How do the constraints fit nicely?
- How do the subtasks work together?
- Come up with lots of observations.
- Write pseudocode.